Ema Lovrić, 18.9.2026.

Assignment 21: Windows system services analysis and management

https://github.com/elovric13/eutech/tree/Windows-system-services-analysis-and-management

1. **Open the Windows Services management tool (services.msc).**

2. **Review the list of services installed on the computer and identify:**

- **Running services.**

The computer has a large number of running services. The most notable categories include system core services, Intel hardware services, VMware virtualization services, Windows Defender security services, audio services, and networking services. The total number of running services observed is approximately 150.

- **Stopped services.**

Several services are installed but not currently running, meaning their Status field is blank. These include ActiveX Installer, App Readiness, Application Identity, Application Layer Gateway Service, and Agent Activation Runtime among others. These are either triggered on demand or not needed at this time.

- **Automatic services.**

Services configured to start automatically with Windows include DHCP Client, DNS Client, Windows Defender Antivirus Service, Windows Defender Firewall, Task Scheduler, Remote Procedure Call (RPC), Windows Audio, WLAN AutoConfig, Print Spooler, SysMain, and many others.

- **Manual services.**

Services configured as Manual start only when triggered by Windows or an application that needs them. Examples include Bluetooth Support Service, Credential Manager, Network List Service, Print Spooler (some variants), Smart Card, and Windows Security Service.

3. **Select 10 Windows services and create a table containing: Service name, Current status. Startup type. Main function.**

Service Name	Status	Startup Type	Main Function
DHCP Client	Running	Automatic	Registers and updates IP addresses and DNS records for the computer
DNS Client	Running	Automatic (Trigger Start)	Resolves domain names to IP addresses and caches results
Windows Defender Firewall	Running	Automatic	Blocks unauthorized network access to protect the computer
Microsoft Defender Antivirus Service	Running	Automatic	Protects against malware and unwanted software
Windows Update (Update Orchestrator)	Running	Automatic (Delayed Start)	Manages downloading and installing Windows updates
Task Scheduler	Running	Automatic (Trigger Start)	Runs scheduled tasks and system maintenance jobs automatically

Windows Audio	Running	Automatic	Manages audio output and input for all Windows applications
Print Spooler	Running	Automatic	Manages print jobs sent to printers
Windows Search	Running	Automatic (Delayed Start)	Indexes files and content for fast search results
Remote Procedure Call (RPC)	Running	Automatic	Core service enabling communication between Windows components and programs

4. **From the selected services, identify which ones are related to:**

Networking.

- DHCP Client – directly responsible for obtaining and maintaining the computer's IP address on the network
- DNS Client – resolves domain names to IP addresses, essential for any internet or network communication
- Windows Defender Firewall – controls inbound and outbound network traffic based on security rules

Security.

- Windows Defender Firewall – protects against unauthorized network access
- Microsoft Defender Antivirus Service – actively scans for and blocks malware and threats
- Remote Procedure Call (RPC) – underpins secure communication between system components

System updates.

- Update Orchestrator Service – manages the full Windows Update cycle, from checking for updates to downloading and installing them
- Task Scheduler – runs scheduled maintenance tasks including update-related background jobs

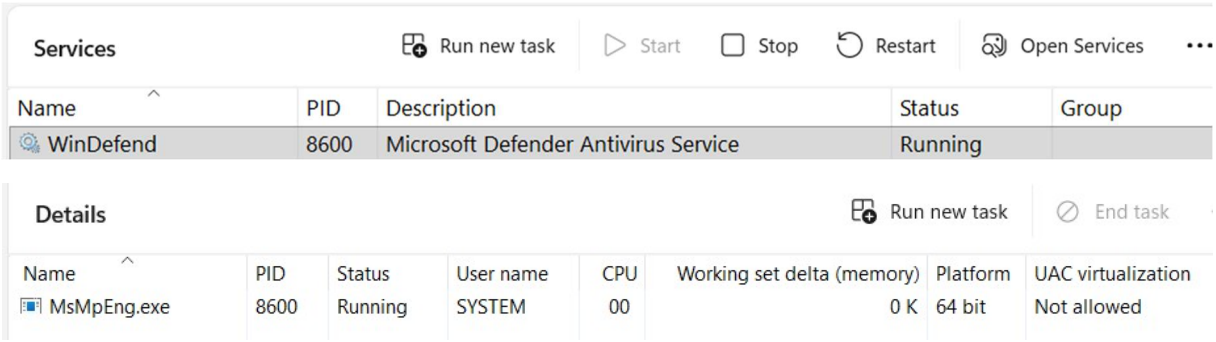
User functionality.

- Windows Audio – without this service no sound works in any application
- Print Spooler – manages all printing jobs, stopping it makes printers unavailable
- Windows Search – provides the file and content search experience used daily from the Start menu and File Explorer
- Task Scheduler – also supports user-facing scheduled tasks such as automated backups or reminders

5. **Select three running services and use Task Manager to identify whether they are associated with active processes.**

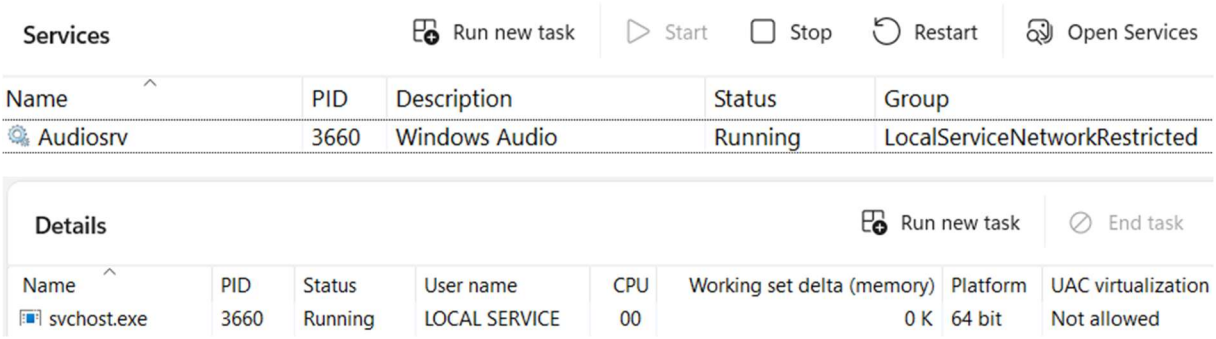
Microsoft Defender Antivirus Service (WinDefend)

In the Services tab, WinDefend was shown as Running with PID 8600. Searching for MsMpEng.exe in the Details tab confirmed the same PID 8600, running under the SYSTEM account. This is the main antivirus scanning process and is one of the most resource-active processes on the system, as also observed in the Resource Monitor analysis.



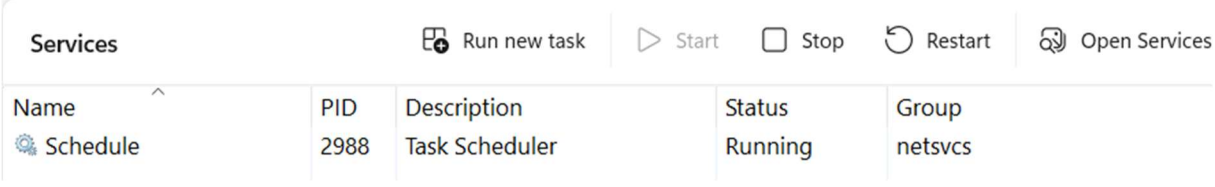
Windows Audio (Audiosrv)

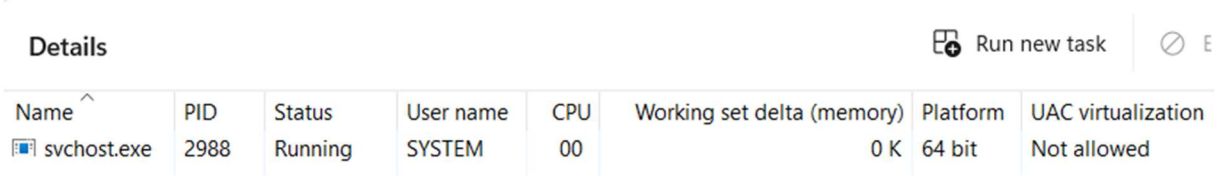
In the Services tab, Audiosrv was shown as Running with PID 3660 under the LocalServiceNetworkRestricted group. The Details tab confirmed svchost.exe with PID 3660 running under LOCAL SERVICE. Windows Audio runs inside svchost.exe, which is standard for many Windows services.



Task Scheduler (Schedule)

In the Services tab, Schedule was shown as Running with PID 2988 in the netsvcs group. The Details tab confirmed svchost.exe with PID 2988 running under SYSTEM. Like Windows Audio, Task Scheduler is hosted inside svchost.exe. The netsvcs group is a common svchost group that hosts several important Windows services simultaneously.





6. **Analyze the different Windows service startup types:**

Automatic.

The service starts immediately when Windows boots, before the user logs in. Examples from this computer: DHCP Client, Windows Defender Firewall, Windows Audio, Print Spooler, Remote Procedure Call.

Automatic (Delayed Start).

The service starts automatically but waits until the core Windows boot process is complete and the system has settled. This reduces boot time by spreading the startup load over a longer period. Examples from this computer: Background Intelligent Transfer Service (BITS), Windows Search, Update Orchestrator Service, Security Center.

Manual.

The service does not start automatically. It only starts when Windows itself, another service, or an application explicitly requests it. Examples from this computer: Bluetooth Support Service, Credential Manager, Network List Service, Smart Card.

Disabled.

The service cannot start at all – not automatically, not manually, not when triggered. It must be re-enabled before it can run. This is used for services that are not needed on a specific machine or that pose a security risk if left available.

7. **Explain when each startup type may be appropriate from a system administration perspective.**

Automatic is appropriate for services that are always needed from the moment the system starts – core networking, security, audio, and RPC services must be available immediately. Delaying them would break functionality that users and other services depend on from the first second of use.

Automatic (Delayed Start) is appropriate for important but non-urgent services like Windows Search and Windows Update. Delaying them reduces the initial boot load and makes the system feel more responsive during startup without affecting functionality. This is the preferred startup type for any non-critical background service that needs to run regularly.

Manual is appropriate for services only needed in specific situations. Bluetooth only needs to run when a Bluetooth device is in use, Smart Card only when a card reader is connected. Keeping these as Manual saves memory and reduces the attack surface.

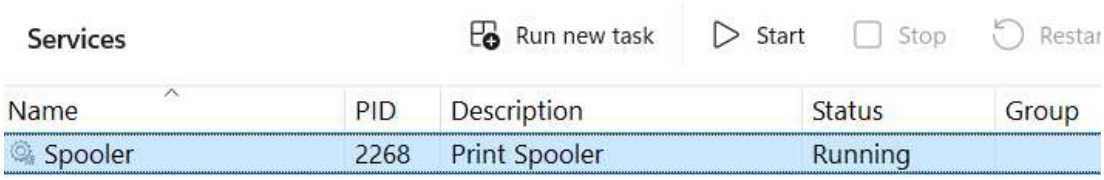
Disabled is appropriate when a service is confirmed unnecessary for the specific computer or poses a security risk. For example, Print Spooler has historically been a target for exploits and should be

disabled on machines that never print. Disabled should be used carefully, because disabling a service that others depend on can cause unexpected failures.

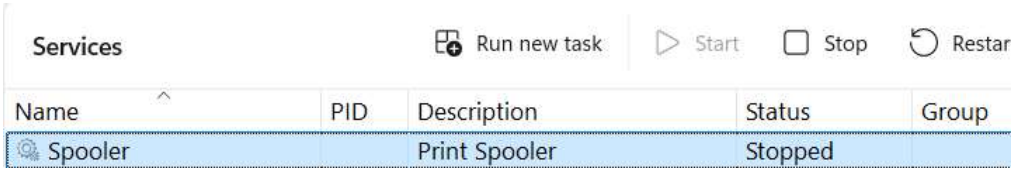
8. **Select one non-essential service that can be safely tested on your computer. Record its initial status, stop it temporarily, observe the result, and start it again. Do not modify antivirus, firewall, Windows Update, networking, authentication, or other critical system services.**

The Print Spooler service was selected for this test as it is a non-essential service for this task – no printing was being performed at the time. It is safe to stop temporarily and has no impact on other running services or system stability.

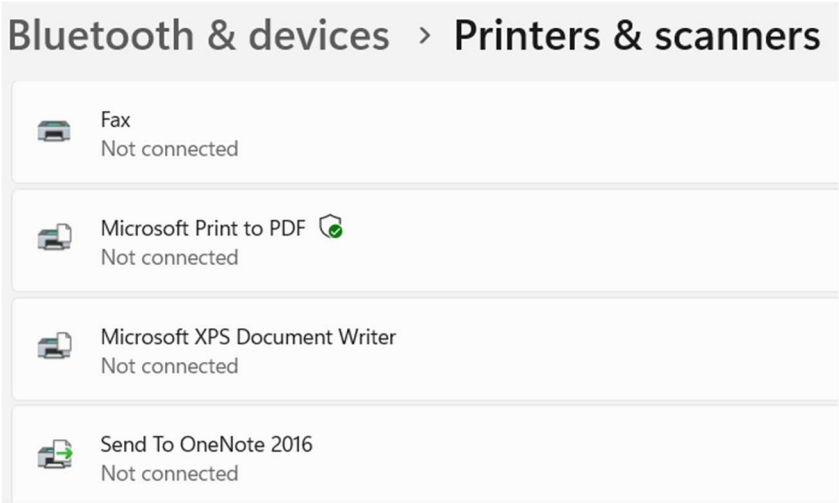
Print Spooler was confirmed as Running with Automatic startup type before the test began.



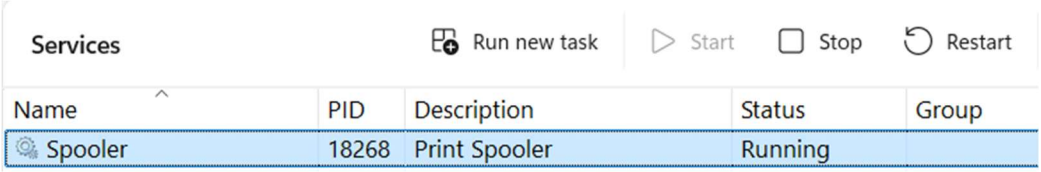
The service was stopped by right-clicking Print Spooler in services.msc and selecting Stop. The status changed from Running to blank (stopped) immediately.



After stopping the service, navigating to Settings → Bluetooth & devices → Printers showed that no printers were available and the system was unable to communicate with any printing devices. This confirms that the Print Spooler service is responsible for all printer communication



The service was restarted by right-clicking Print Spooler and selecting Start. The status returned to Running immediately and printer functionality was restored.

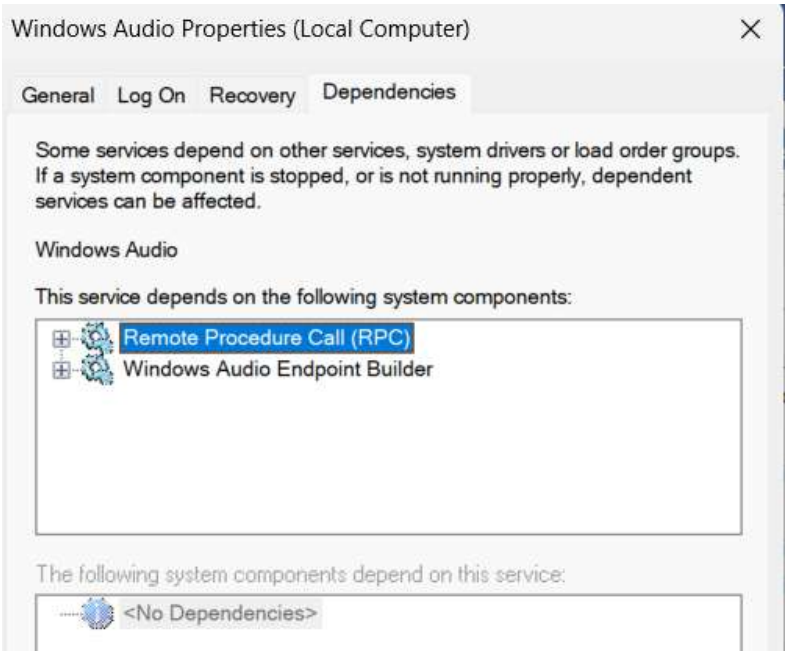


This test demonstrates that even a service marked as non-critical can have an immediate and visible impact on user functionality when stopped. It also demonstrates how quickly services can be stopped and restarted through services.msc without requiring a system reboot, which is a key advantage of the Windows services management tool for system administrators.

9. **Review the properties of five services and identify any dependencies between services.**

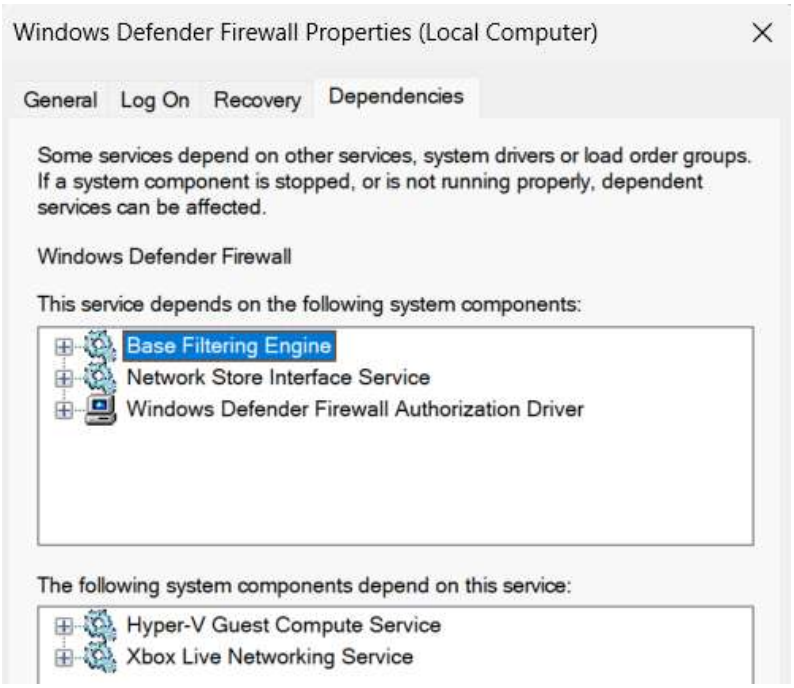
Windows Audio

- Depends on: Remote Procedure Call (RPC), Windows Audio Endpoint Builder
- Required by: No dependencies
- Windows Audio relies on RPC for communication and on the Endpoint Builder service to manage audio devices. No other services depend on it, so stopping it only affects sound output/input directly, with no cascading effect on other services.



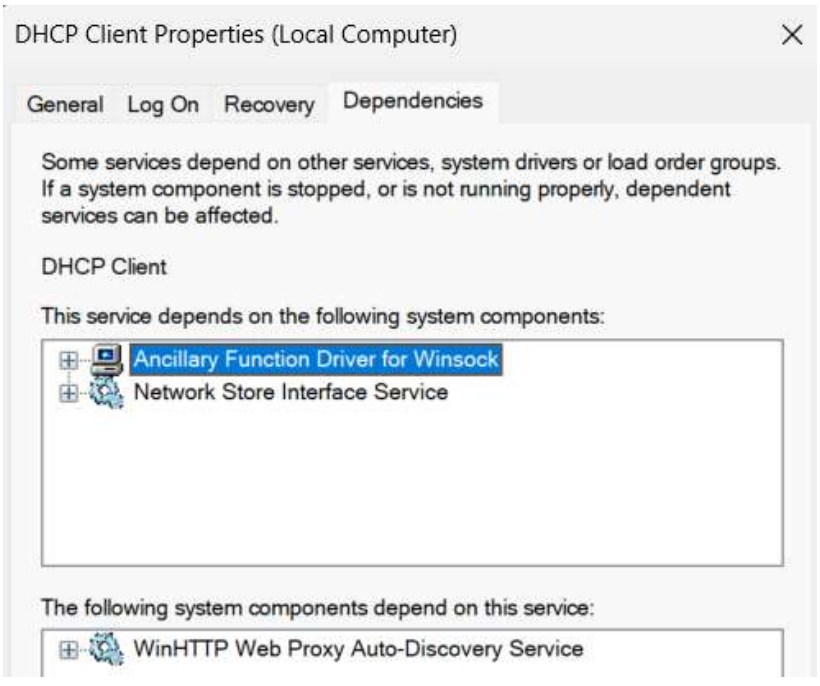
Windows Defender Firewall

- Depends on: Base Filtering Engine, Network Store Interface Service, Windows Defender Firewall Authorization Driver
- Required by: Hyper-V Guest Compute Service, Xbox Live Networking Service
- This service has both significant dependencies and dependents, making it a good example of a service positioned centrally in the service chain. If Base Filtering Engine fails, the firewall cannot start; if the firewall itself is stopped, both Hyper-V networking and Xbox Live networking features are affected.



DHCP Client

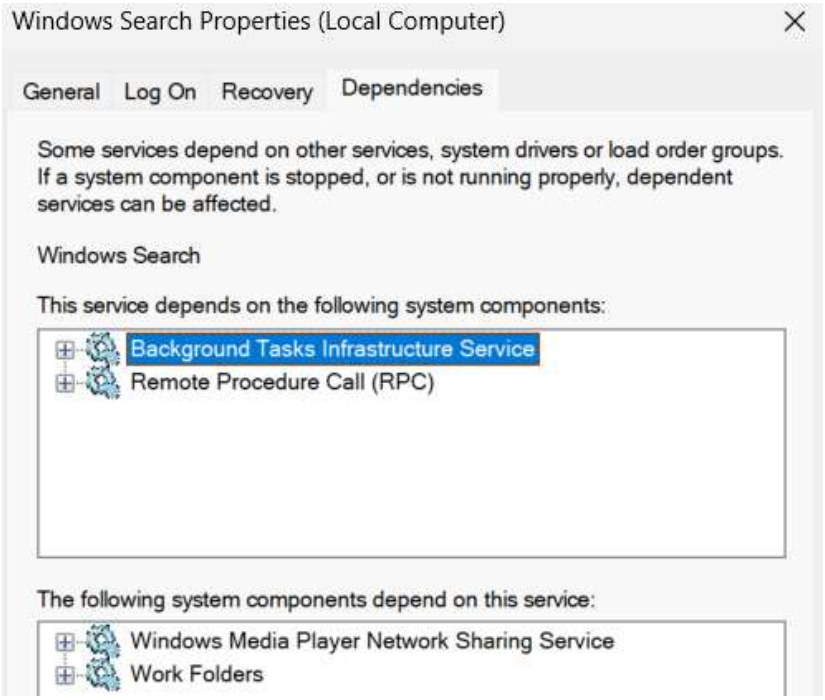
- Depends on: Ancillary Function Driver for Winsock, Network Store Interface Service
- Required by: WinHTTP Web Proxy Auto-Discovery Service
- DHCP Client is responsible for obtaining IP addresses automatically. It depends on core networking drivers, and the WinHTTP Proxy Auto-Discovery service depends on it, so a failure here can disrupt both network addressing and proxy detection.



Windows Search

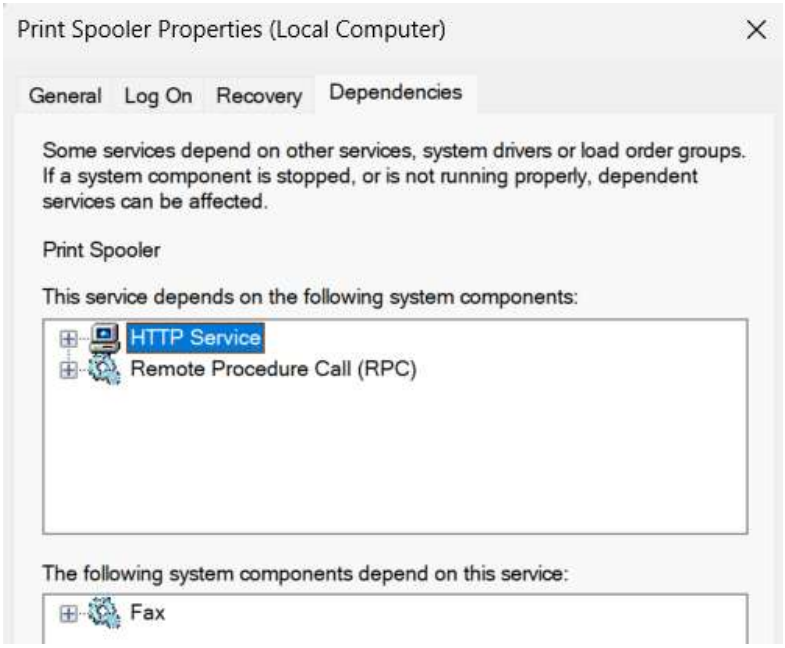
- Depends on: Background Tasks Infrastructure Service, Remote Procedure Call (RPC)
- Required by: Windows Media Player Network Sharing Service, Work Folders

- Windows Search needs background task management and RPC to index and run. Two features, media sharing and Work Folders, rely on it, showing how a background indexing service can quietly affect unrelated features if stopped.



Print Spooler

- Depends on: HTTP Service, Remote Procedure Call (RPC)
- Required by: Fax
- Print Spooler cannot function without RPC, which handles inter-process communication, and HTTP Service, used for certain print protocols. In turn, the Fax service depends on Print Spooler, meaning stopping Print Spooler would also disable faxing.



10. **Identify three possible problems that could occur when an important Windows service fails or does not start correctly.**

1. Loss of network connectivity

If a core networking service such as DHCP Client or DNS Client fails to start, the computer loses the ability to obtain an IP address or resolve domain names. In practice this means the user cannot access the internet, shared network drives, or any network resource, even if the physical network connection is working perfectly. This is one of the most common and disruptive service failures in enterprise environments because it affects every application that requires network access.

2. Security vulnerabilities

If a security-related service such as Windows Defender Firewall or Microsoft Defender Antivirus fails to start, the computer is left unprotected. The firewall not running means inbound and outbound traffic is no longer filtered, exposing the system to network-based attacks. The antivirus not running means malware can execute without being detected or blocked. These failures are particularly dangerous because the user may not notice immediately, the computer continues to function normally in other respects while the protection is gone.

3. System instability and application failures

If a core infrastructure service such as Remote Procedure Call (RPC) or Windows Management Instrumentation (WMI) fails, a large number of other services and applications that depend on them will also fail. RPC is a dependency for dozens of Windows services, if it goes down, services that rely on it cannot start, which can cascade into widespread system instability. In extreme cases this can make the system unbootable or require a repair installation.

11. **Explain how a system administrator could use the Services management tool when troubleshooting Windows problems.**

When a user reports that a feature is not working, printing, audio, network access, Windows Update, the administrator opens services.msc and checks whether the relevant service is running. A stopped service that should be running is often the entire explanation, and restarting it resolves the issue immediately.

When restarting fails, services.msc displays the error generated by the failed start attempt, giving the administrator a starting point for deeper investigation. The Dependencies tab is particularly useful here. If a service fails because another service it depends on is not running, the administrator can trace the dependency chain and fix the root cause rather than the symptom.

Startup type is also a quick diagnostic, if a service that should be Automatic has been changed to Manual or Disabled by an application installer or group policy, services.msc makes this immediately visible and correctable.

Finally, services.msc allows administrators to stop and restart services without rebooting the machine, which in a production environment where uptime matters is a significant operational advantage.